

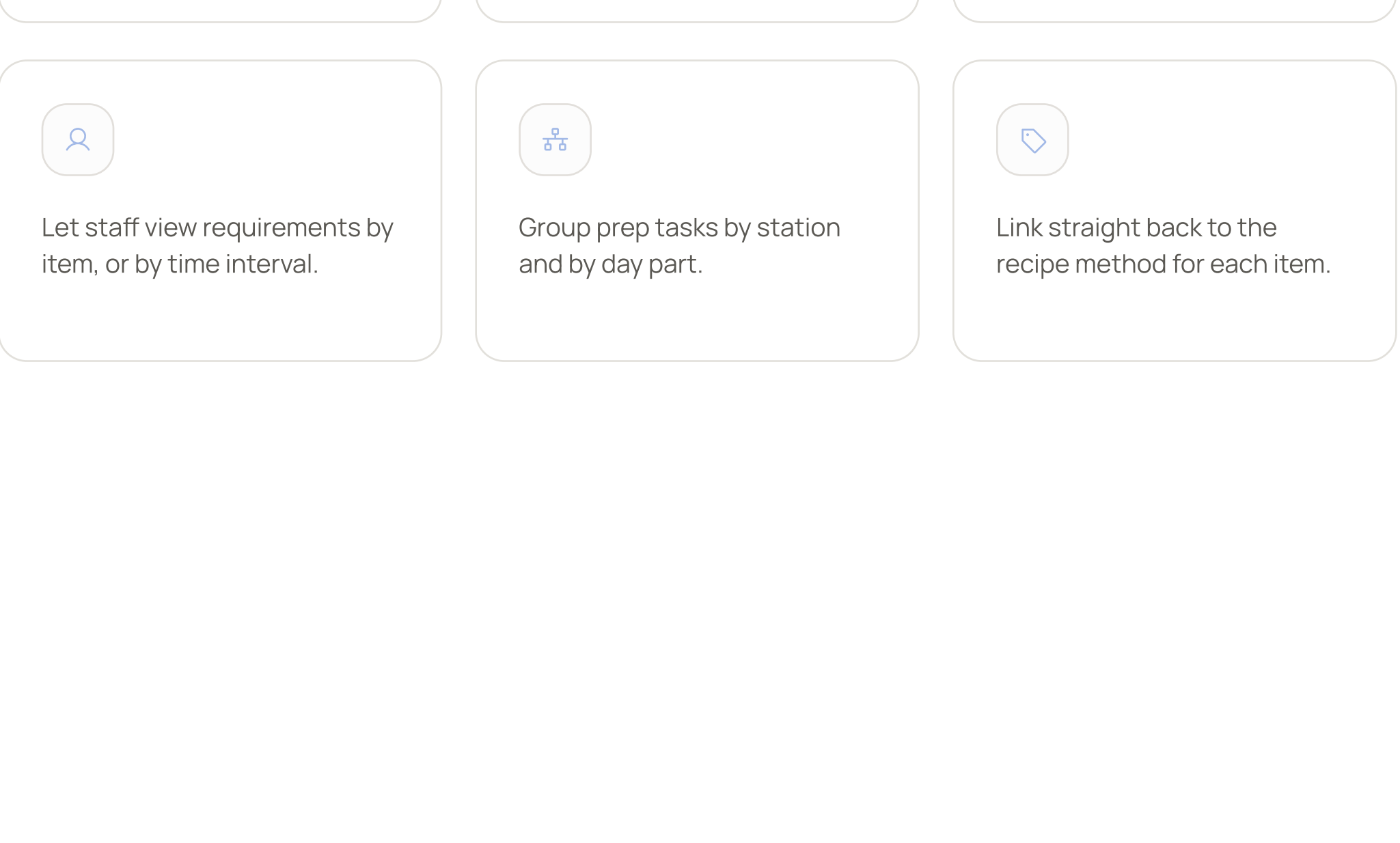
Prep Prediction

A forecasting engine built from zero inside a live enterprise product – turning six weeks of usage history and same-day sales into a **prep**, **thaw**, and **cook plan** kitchens can actually run on.

SynergySuite is a back-office operations platform for multi-location restaurant and hospitality businesses. As part of the **core product team**, I worked on turning complex operational requirements into **clear, usable** product experiences.

Before Prep Prediction, kitchens planned prep, thawing, and cooking from memory and paper par sheets: no forecast, no visibility into what the day's sales would actually demand. Competitors like Perfect Tech already shipped this as a baseline feature. We didn't.

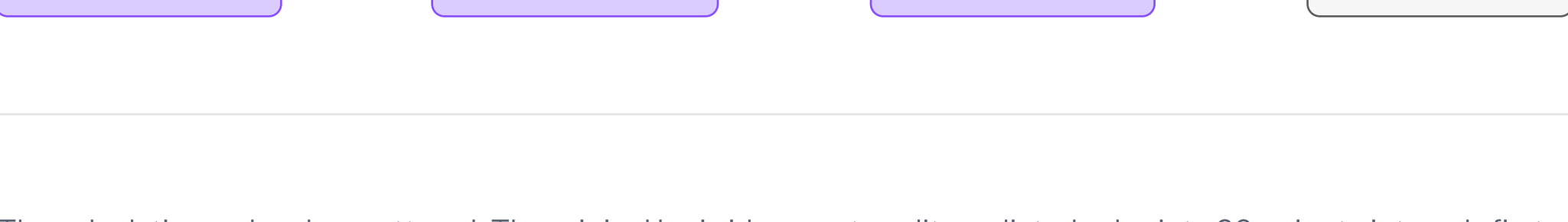
Core product requirements



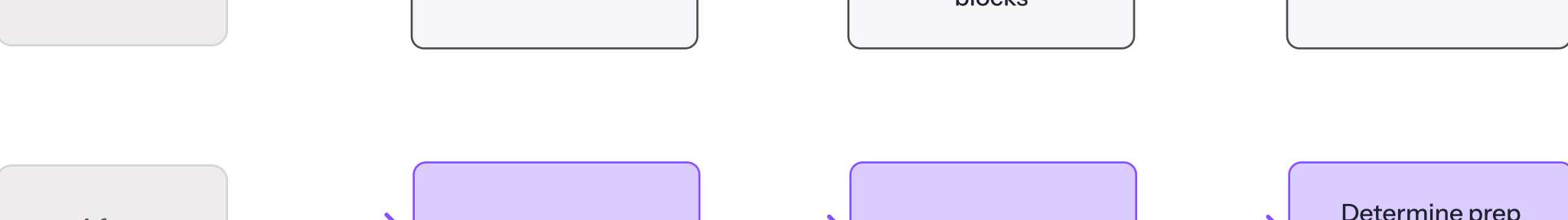
02 - THE CALCULATION

Two prediction methods, one prep plan

Trend-based prediction uses six weeks of item-level usage history to estimate what tomorrow needs. **Shelf life, prep time, and prep frequency** then determine when that quantity should appear, so items are not scheduled too early or placed in the wrong interval.

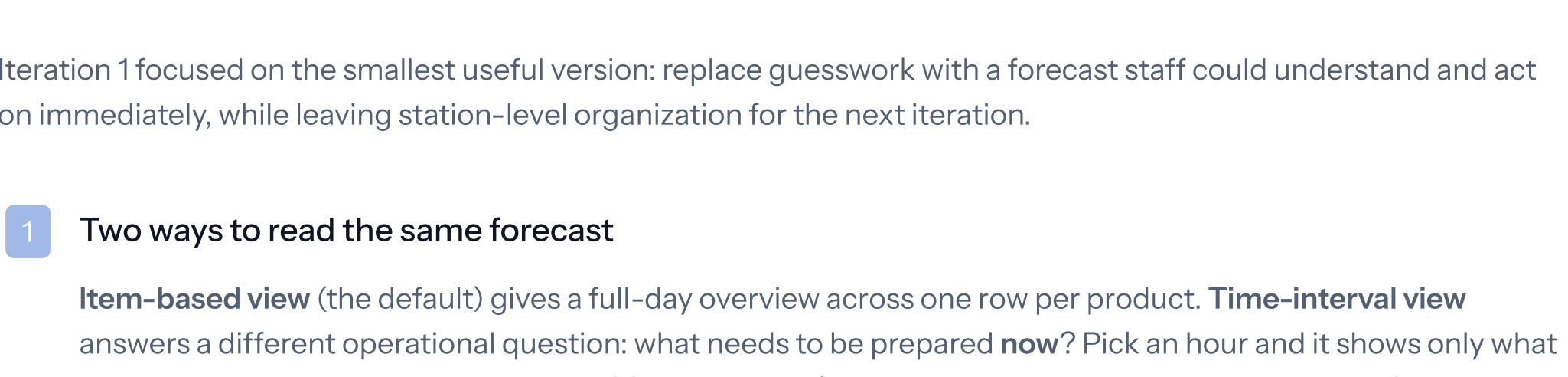


PMIX' forecast prediction ties prep demand to the **sales forecast** itself, which is a separate feature. **Prep items** are linked to the products they support – for example, a patty to a burger – so projected sales scale the required quantity. If \$1,000 in sales usually produces 10 burgers, a \$2,000 forecast predicts 20.



The calculation order also mattered. The original logic idea was to split predicted sales into 30-minute intervals first, and only then apply prep-specific variables such as prep time, prep frequency, and minimum prep amounts. The problem was that those variables can change when a product actually needs to be prepared.

By applying them after the sales had already been divided into time slots, some quantities could be assigned to an interval that did not reflect the real prep requirement. Reversing that order – **applying the prep variables first**, then determining the interval – produced predictions that aligned more closely with actual client usage.



1. PMIX stands for Product Mix – a breakdown of exactly what was sold, and how much of each item, over a given period (e.g., 10 burgers, 15 pizzas), typically pulled from POS/COGS transaction data.

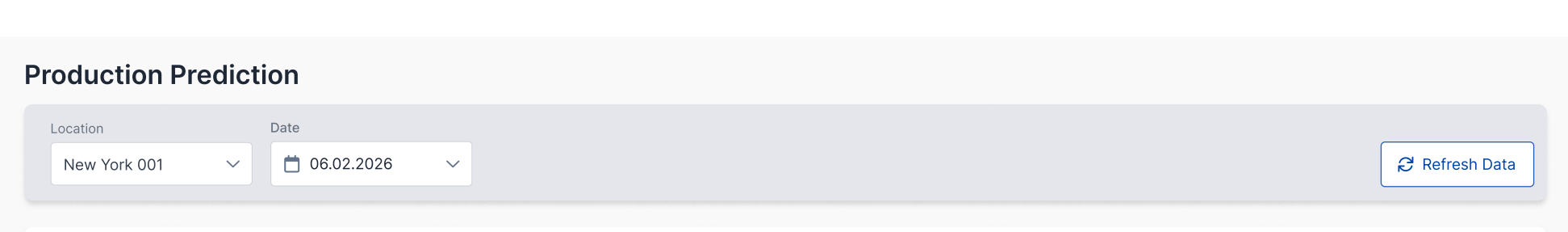
03 - ITERATION 1

Making the forecast usable on shift

Iteration 1 focused on the smallest useful version: replace guesswork with a forecast staff could understand and act on immediately, while leaving station-level organization for the next iteration.

1 Two ways to read the same forecast

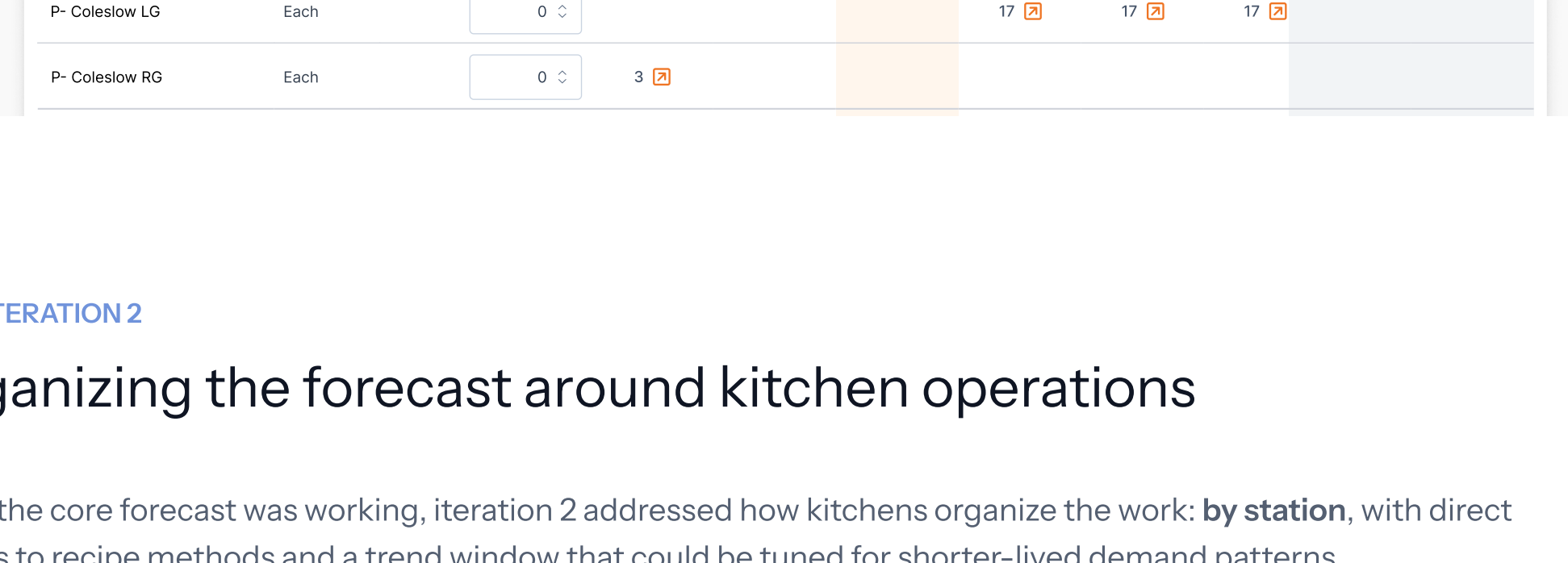
Item-based view (the default) gives a full-day overview across one row per product. **Time-interval view** answers a different operational question: what needs to be prepared **now**? Pick an hour and it shows only what is due inside that window, split into two 30-minute blocks, with zero-quantity items hidden. Every filter, print, and export respects whichever view is active.



2 Splitting the day into parts

A day-part filter lets staff focus on and print only what's relevant to their shift. All day parts are selected by default, and the filter only controls what's visible – it doesn't change the underlying prep intervals.

Day parts are configured through the Day Parts Manager in Settings, a separate feature we designed in parallel and integrated into Prep Prediction.



04 - ITERATION 2

Organizing the forecast around kitchen operations

Once the core forecast was working, iteration 2 addressed how kitchens organize the work: **by station**, with direct access to recipe methods and a trend window that could be tuned for shorter-lived demand patterns.

1 Organizing prep by station

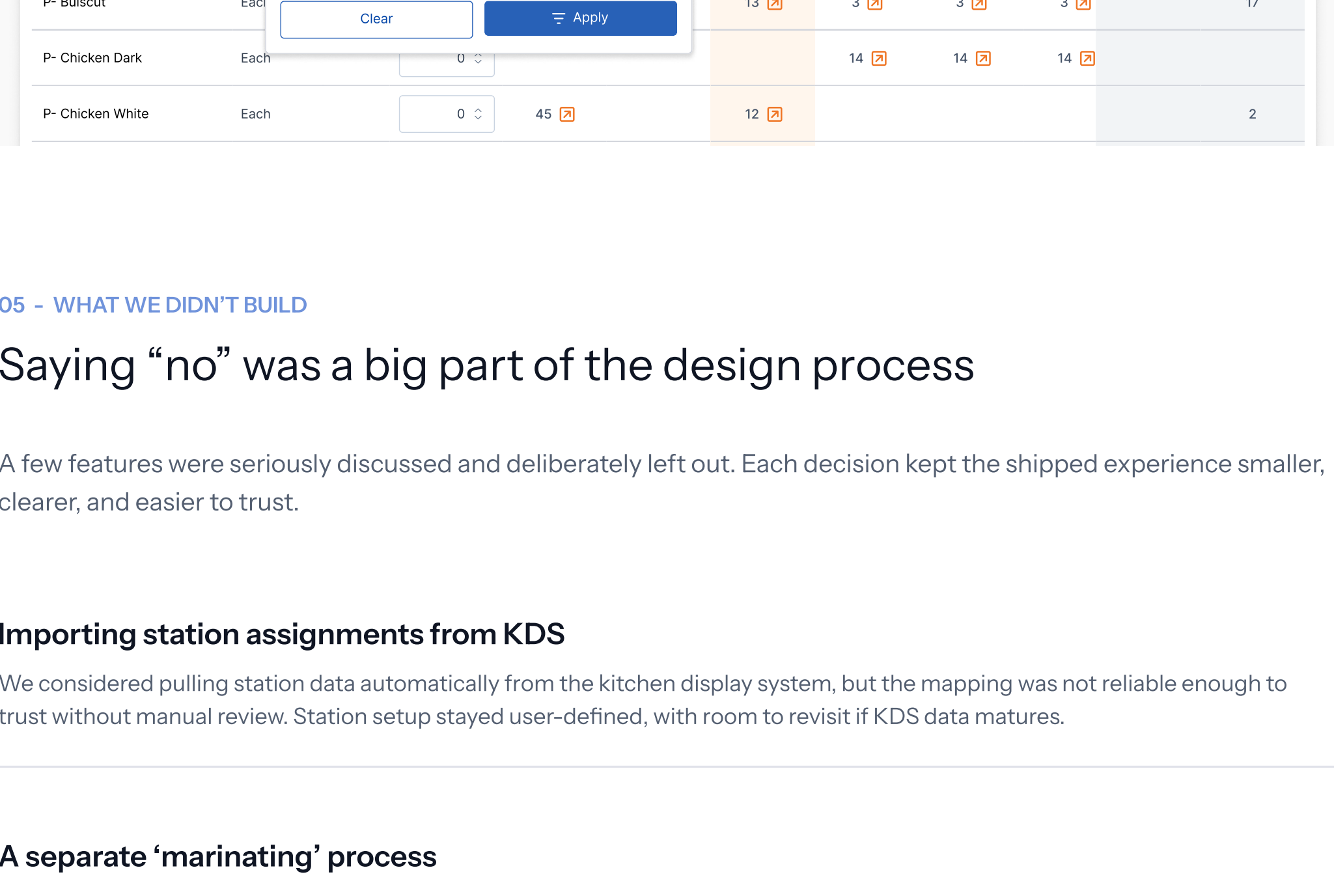
A new Station Setup page (gated behind its own permission) also designed in parallel with Prep Prediction, lets an operator name stations and assign each prep item to exactly one. Items without a station fall into an 'Unassigned' group automatically, which is itself a filterable option on the prep sheet. Deleting a station releases its items back into 'Unassigned' rather than losing the assignment.

2 One click to the recipe

Any prep item with a recipe attached gets a 'View Recipe Method' link right next to it on the prep sheet, opening the method in a pop-up without leaving the page. If no recipe exists, the action stays hidden.

3 Tuning the trend window

Trend-based prediction defaults to six weeks of history – too long for a promo that only ran for two. A configurable setting lets that window shrink per use case, so short-lived items get a forecast built from data that actually reflects them.



05 - WHAT WE DIDN'T BUILD

Saying “no” was a big part of the design process

A few features were seriously discussed and deliberately left out. Each decision kept the shipped experience smaller, clearer, and easier to trust.

Importing station assignments from KDS

We considered pulling station data automatically from the kitchen display system, but the mapping was not reliable enough to trust without manual review. Station setup stayed user-defined, with room to revisit if KDS data matures.

A separate 'marinating' process

Marinating turned out to be prep by another name. Once thaw and prep times could be chained – thaw, then prep, then cook – there was no case a standalone marinating step could handle that reusing the prep process could not.

Hand-editing predictions from the sheet

Letting staff edit predicted quantities directly raised questions we were not ready to answer – permissions, audit trail, and whether edits should feed back into future forecasts. We solved buffers at the product level instead and left this for later.

Multiple prep periods per location

One client's case looked like it needed two daily prep windows – until we realized the second window was for tomorrow's dough, not today's service. The real problem was next-day prep, not concurrent periods, so we did not build a feature nobody needed.

What I learned from building the feature and what would I do differently

The most valuable moment on this project was not a screen. It was reordering the calculation and testing it against a real client's data. A good reminder that prep prediction is a math problem before it is a UI problem, and that the two rarely get solved in the same meeting.

Three things stayed with me:

- validate the logic, not just the interface;
- deciding what not to build is part of design;
- keeping iteration 1 focused protected the scope so iteration 2 could still ship on schedule.

What I would do differently: I'd reconsider the order of the Thaw, Prep, and Cook tabs so they follow the actual sequence of work in the kitchen. Since an item needs to be thawed before it can be prepped, and prepped before it can be cooked, presenting the tabs in that same order would better reflect the operational workflow and make the progression between stages more intuitive.